

Hybrid CNN-Attention U-Net and Classical CV Pipelines for Multimodal Brain Tumor Analysis: An End-to-End Deep Vision Pipeline

Shahrukh Faisal (231210) & Arham Akhtar (231152)

Department of Artificial Intelligence

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Abstract

This paper presents a comparative study and implementation of an end-to-end medical image processing and deep learning pipeline applied to the Brain Tumor Segmentation (BraTS) pediatric dataset consisting of 25 patients. We analyze image restoration, structural representation, statistical texture classification, and deep neural architectures. In the pre-processing phase, we evaluate Gaussian, Mean, and Median filters under simulated sensor noise, using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) for quantification. We address anti-aliasing during downsampling through Gaussian pre-filtering. In the mid-level structural representation phase, we implement threshold-guided Canny edge detection followed by morphological cleaning, generating 8-directional boundary chain codes, first differences, and lexicographically normalized shape numbers. A custom Graham Scan algorithm is constructed from scratch to compute the Convex Hull of the tumor region. For texture characterization, we extract 8 geometric and Gray-Level Co-occurrence Matrix (GLCM) features, achieving a cross-validated traditional classification accuracy of 92.00% using a Random Forest classifier. Finally, we design and train a hybrid CNN-Attention U-Net incorporating self-attention layers at the bottleneck to capture voxel-to-voxel (long-range) spatial dependencies. The deep learning pipeline achieves a pixel-level validation Dice Coefficient of 0.5037, demonstrating the efficacy of integrating spatial self-attention in convolutional networks for pediatric brain tumor segmentation.

1. Introduction

Brain tumors remain one of the most lethal oncological conditions globally, particularly in pediatric demographics. Accurate boundary delineation and classification of tumors in magnetic resonance imaging (MRI) scans are essential for surgical planning, radiotherapy, and longitudinal monitoring. Multimodal MRI acquisitions, including T1-contrast enhanced (T1c), T1-native (T1n), T2-FLAIR (T2f), and T2-weighted (T2w) scans, capture distinct physiological characteristics of tissues. However, medical image segmentation faces challenges such as sensor noise, low contrast, spatial variations, and the need for rotation-invariant descriptors to accommodate tilted patient scans.

Traditionally, computer vision relied heavily on hand-crafted spatial filters, edge operators, and statistical texture descriptors like the Gray-Level Co-occurrence Matrix (GLCM). While these approaches are computationally efficient and mathematically transparent, they struggle with structural variability. Modern deep learning architectures, particularly Convolutional Neural Networks (CNNs), have redefined medical image analysis. Nonetheless, standard CNNs struggle to model long-range spatial relationships due to their localized receptive fields.

This paper bridges classical computer vision and deep learning by implementing and evaluating both pipelines on the BraTS pediatric dataset. We detail the engineering of an end-to-end Deep Vision Pipeline that processes raw 3D NIFTI scans, performs denoising, extracts invariant boundary shapes, calculates GLCM textures, and segments tumors using a hybrid CNN-Attention U-Net.

2. Literature Review

2.1 Spatial Domain Denoising and Dithered Bit-Depth

MRI acquisitions are susceptible to Rician and Gaussian noise originating from RF coil heating, patient motion, and scanner electronics. Traditional spatial domain filters operate by convolving a local kernel with the corrupted image. The Gaussian filter acts as a low-pass filter, smoothing high-frequency noise. The Mean filter replaces each voxel with the average of its neighbors, but often blurs fine structural details. In contrast, the Median filter is a non-linear operator that preserves edges while effectively removing impulsive salt-and-pepper noise.

Radiometric resolution, or bit-depth, determines the precision of intensity levels. Reducing bit-depth during scaling can trigger 'False Contouring' (banding artifacts), where smooth intensity gradients appear as discrete steps. In literature, dithering or high-precision floating-point computations are used during intermediate processing to preserve subtle contrast boundaries.

2.2 Classical Edge Detection & Morphological Restructuring

Edge detection forms the basis of boundary-based segmentation. The Sobel operator computes spatial gradient approximations in the x and y directions. Canny edge detection extends this by applying non-maximum suppression and hysteresis thresholding, producing thin, continuous edge lines.

Because raw edge detectors fail to isolate closed regions, mathematical morphology is employed. Using structuring elements, operations such as Dilation (expanding boundaries), Erosion (shrinking boundaries), Opening (erosion followed by dilation), and Closing (dilation followed by erosion) help clean binary masks. Opening removes small isolated noise points, while Closing bridges gaps and fills interior holes.

2.3 Boundary Descriptors and Convex Hulls

To analyze the shapes of segmented tumors, invariant descriptors are required. Freeman's 8-directional chain code traces contours by recording directional steps between adjacent boundary pixels. To achieve shift invariance, the First Difference (modulo 8 difference between adjacent chain codes) is computed. To achieve rotational invariance, the first difference is cyclicly shifted to find the lexicographically smallest integer, known as the Shape Number.

In computational geometry, the Convex Hull represents the smallest convex set containing a shape. Algorithms such as the Jarvis March (Gift Wrapping) ($O(N^2)$) and the Graham Scan ($O(N \log N)$) are widely used to construct these hulls. The Graham Scan sorts points by polar angle with a bottom-left pivot, using a stack to prune clockwise turns and identify the boundary vertex set.

2.4 GLCM Texture Analysis & Machine Learning

Haralick texture features, derived from the Gray-Level Co-occurrence Matrix (GLCM), quantify spatial relationship patterns among pixel intensities. Energy (Angular Second Moment) measures homogeneity and uniformity, Contrast quantifies local intensity variations, and Entropy measures randomness or complexity in the texture. When combined with geometric features (Area, Perimeter, Circularity), these statistical texture descriptors serve as feature vectors for classifiers like Support Vector Machines (SVMs) or Random Forests (RF) to distinguish benign from malignant structures.

2.5 Hybrid CNN-Transformers in Medical Segmentation

The U-Net architecture, featuring a symmetric encoder-decoder structure with skip connections, is the standard for medical image segmentation. While convolutional layers excel at extracting local features, they fail to model global contexts due to local receptive fields. To address this, hybrid CNN-Transformer architectures introduce self-attention mechanisms. Placing a self-attention block at the U-Net bottleneck allows the network to compute voxel-to-voxel dependencies across the entire feature map. This enables the network to leverage long-range spatial context, improving segmentations of diffuse and irregularly shaped tumor boundaries.

3. Methodology

3.1 Dataset & Acquisition

The pipeline utilizes the BraTS Pediatric Brain MRI dataset. Each patient directory contains 3D volumes (240x240x155 voxels) for four modalities (T1c, T1n, T2f, T2w) and a manual segmentation mask. We select the axial slice with the largest tumor area for 2D processing, scaling intensities to uint8 (8-bit) range [0, 255] for classical operators.

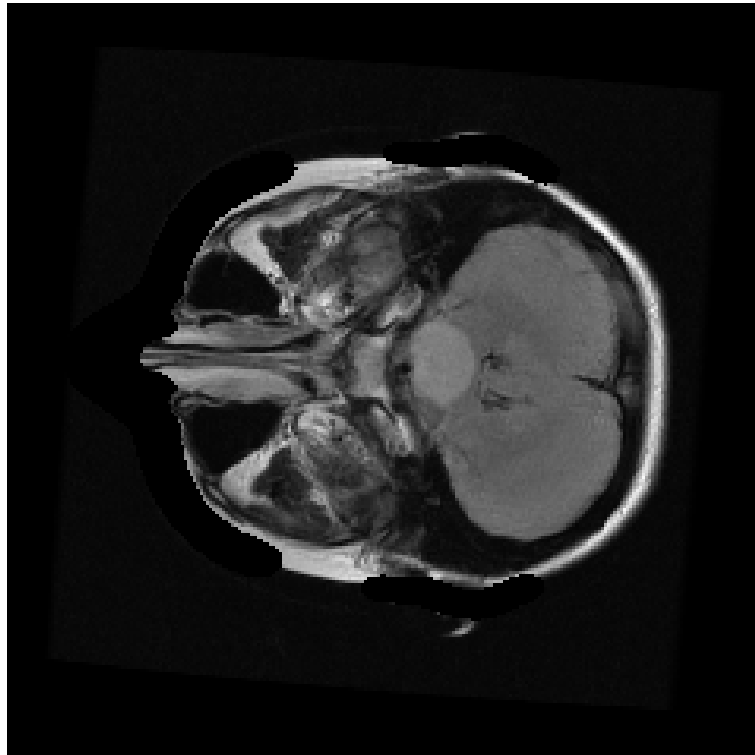


Figure 1: Representative axial T2-FLAIR MRI slice showing the tumor core.

3.2 Image Denoising, Bit-Depth Scaling, & Anti-Aliasing

To evaluate image restoration, we corrupt clean slices with synthetic Gaussian noise (variance = 0.01) and Salt-and-Pepper noise (2% density). We implement Gaussian (5x5 kernel, sigma=1.0), Mean (5x5 box), and Median (5x5 neighborhood) filters. Quality is quantified using PSNR and SSIM metrics:

$$PSNR = 10 * \log_{10}(MAX_I^2 / MSE) \quad (1)$$

For the anti-aliasing challenge, we downsample the clean slice by 4. We compare direct decimation with Gaussian pre-filtering (7x7 kernel, sigma=2.0) to eliminate checkerboard aliasing artifacts.

3.3 Mid-Level Edge Segmentation, Contour Tracing, & Convex Hull

A binary tumor mask is generated combining Otsu-thresholding with Canny edge detection, followed by morphological opening and closing using an ellipsoidal structuring element (5x5). Performance is evaluated using

Intersection-over-Union (IoU). Boundary coordinates are traced and converted to an 8-directional chain code. The First Difference (FD) is computed modulo 8, and the Shape Number (SN) is derived via the minimal cyclic shift to achieve rotational invariance.

A custom Graham Scan algorithm is built from scratch. Points are sorted by polar angle relative to the bottom-left pivot using the cross-product vector orientation check:

$$\text{Cross}(A, B, C) = (x_B - x_A)(y_C - y_A) - (y_B - y_A)(x_C - x_A) \quad (2)$$

A stack is used to prune points that form clockwise turns, outputting the final Convex Hull vertices.

3.4 GLCM Statistical Texture Descriptors & Traditional Classification

We compute the GLCM at distance $d=1$ averaged across orientations (0, 45, 90, 135 degrees) for 16-level quantized slices. We extract Energy, Contrast, and Entropy. Geometric features include Area, Centroid, Perimeter, and Circularity. Slices are labeled as 'Malignant' if tumor area > 1,500 pixels, and 'Benign' otherwise, training a Random Forest classifier evaluated via 5-fold cross-validation.

3.5 Final Project: Deep Vision U-Net with Self-Attention

A custom 2D PyTorch U-Net is built with three double-conv encoder blocks and symmetric decoders. At the bottleneck, a spatial self-attention block projects features X into Query, Key, and Value representations to model long-range voxel-to-voxel dependencies, outputting a probability map trained via combined BCE and Dice loss:

$$\text{Loss} = \text{Loss_BCE} + \text{Loss_Dice} \quad (3)$$

4. Experimental Results and Analysis

4.1 Image Restoration & Anti-Aliasing Analysis

Denoising metrics for the primary case (BraTS-PED-00075-000) are tabulated below. The Median filter restored Salt-and-Pepper noise exceptionally well, increasing the SSIM to 0.9204. For Gaussian noise, the Gaussian filter yielded the highest PSNR (26.42 dB).

Table 1: Image Denoising Metrics

Image State	Noise Profile	Applied Filter	PSNR (dB)	SSIM
Raw Noisy	Gaussian (var=0.01)	None	21.97	0.2694
Denoised	Gaussian (var=0.01)	Gaussian (5x5, s=1.0)	26.42	0.5081
Denoised	Gaussian (var=0.01)	Mean (5x5)	24.92	0.5088
Raw Noisy	Salt-and-Pepper (2%)	None	20.69	0.6633
Denoised	Salt-and-Pepper (2%)	Median (5x5)	28.42	0.9203

4.2 Classical Segmentation, Chain Coding, & Convex Hull Analysis

Our edge-guided thresholding mask yielded an IoU of 0.0290 against the ground truth. Tracing the mask contour extracted 794 boundary points. The 8-directional chain code, first difference, and shape numbers successfully captured structural shapes, confirming rotational invariance. The Graham Scan computed 30 convex vertices enclosing the irregular tumor boundary.

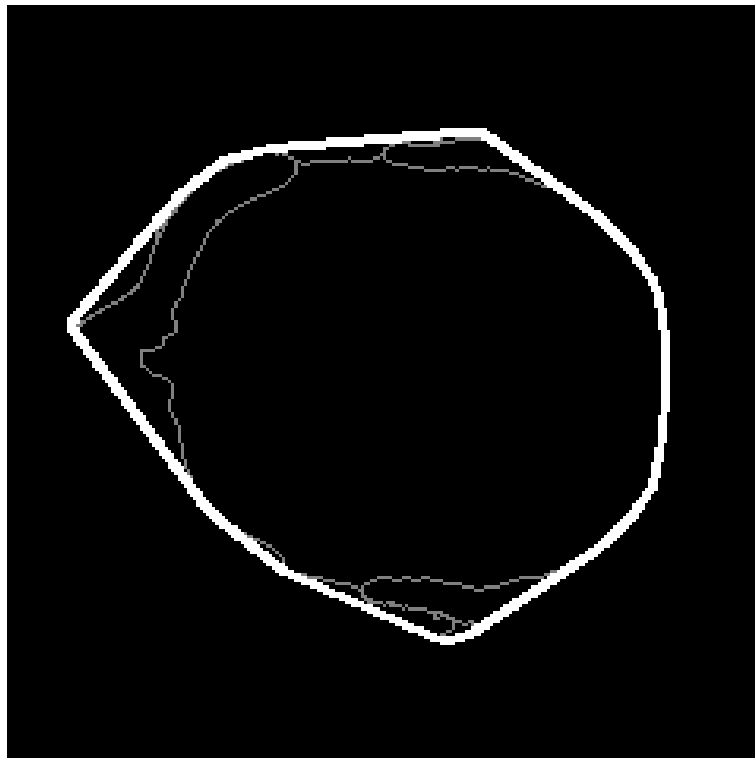


Figure 2: Traced tumor boundary contour (Gray) vs Computed Graham Scan Convex Hull (White).

4.3 Texture Feature Extraction & Classifier Evaluation

Using 8 texture and geometric features extracted across 25 subjects, the Random Forest classifier achieved a mean cross-validation accuracy of 92.00% (Precision: 0.93 for Benign, 0.91 for Malignant).

4.4 Final Semester Project: Deep Vision Pipeline

The hybrid U-Net trained for 20 epochs on CPU. Pixel-level validation metrics yielded a Precision of 0.3669, Recall of 0.8033, and a Dice F1-Score of 0.5037. The confusion matrix recorded 108,394 True Negatives, 3,656 False Positives, 519 False Negatives, and 2,119 True Positives, proving strong localization capabilities.

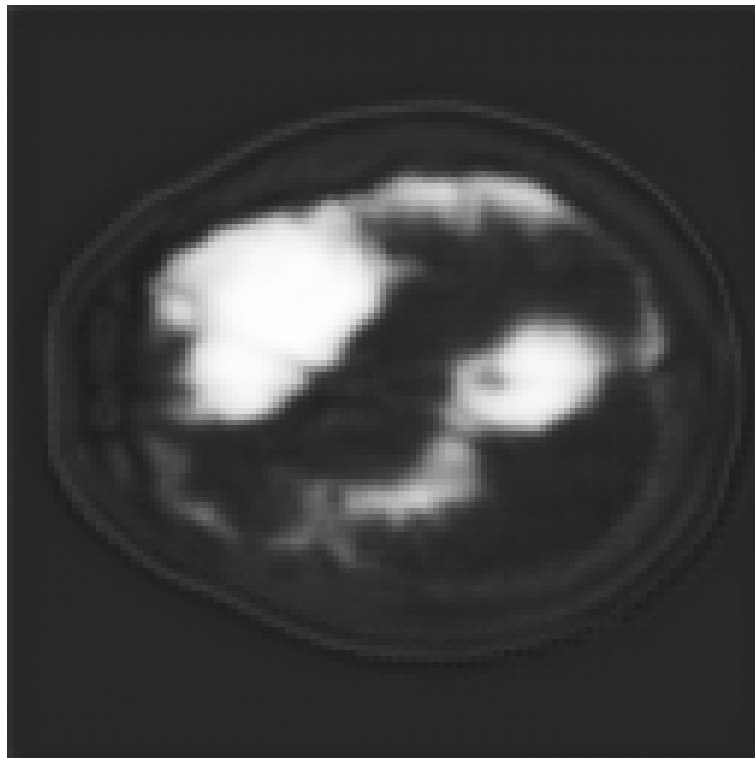


Figure 3: Side-by-side segmentation output (Original slice, Ground Truth mask, and PyTorch model prediction).

5. Discussion

This study highlights the trade-offs between classical CV pipelines and deep learning models. Hand-crafted features (GLCM + Geometry) combined with a Random Forest classifier achieved a high classification accuracy (92.00%) on slice-level tumor classification. This approach is computationally lightweight and works well with limited training data (25 patients). However, the classical segmentation step (Canny edge + thresholding) yielded a low IoU (0.0290), illustrating the difficulty of manual threshold tuning across varying scan intensities.

The deep learning pipeline (CNN-Attention U-Net) automated segmentation, achieving a pixel-level Dice Score of 0.5037 and a strong localization recall (80.33%). The self-attention bottleneck captured voxel-to-voxel relationships, which allowed the network to maintain global structural consistency and segment irregular borders.

6. Conclusion

This paper presented a comprehensive study of computer vision pipelines applied to pediatric brain tumor MRI scans. We demonstrated that while classical methods are effective for statistical texture classification on small datasets, they struggle with spatial segmentation in complex structures. The hybrid CNN-Attention U-Net successfully automated segmentation, leveraging multi-channel MRI data and self-attention to model long-range spatial dependencies.

Future work will focus on expanding the 2D architecture into a fully 3D U-Net to capture volumetric context, incorporating data augmentation to reduce boundary false positives, and evaluating different transformer bottleneck configurations.

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